

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Bachelor of Physiotherapy (B.P.T)

Semester – II Examination – JUNE 2011

PY102 FUNDAMENTALS OF BIOMEDICAL PHYSICS

Date: 16.06.11, Thursday

Time: 10.00a.m to 10.20a.m

Maximum Marks: 10

Instructions:

1. There are two parts in the paper. **PART –I** contains 10 Questions and **PART – II** contains 4 Questions.
2. **PART –I** should be answered in MCQ 'S question paper & **PART – II** should be answered in two separate answer books which has two sections.
3. Marks on the right side suggest the total marks of that question.
4. Maximum time allotted for **PART – I** is 20 min. Question paper for **PART – II** will be given at the table when the answer sheet for **PART – I** is returned. However, Candidates who complete **PART – I** earlier than 20 min, can collect main question paper for **PART –II** immediately after handing over the **PART-I** question paper.

PART-I

Multiple Choice Questions: (MCQs)
(Encircle the correct answer)

(1 × 10 = 10)

1. The refractive index of the core and materials of fiber optic are 1.54 and 1.5 respectively than what is the value of Numerical Aperture?
 - a. 0.526
 - b. 0.2986
 - c. 0.3487
 - d. 0.4283
2. Velocity of sound in air is
 - a. 323.8m/s
 - b. 332.8cm/s
 - c. 330.8m/s
 - d. 332.8m/s
3. What is the Newton's second law
 - a. $E=hf$
 - b. $F=m \cdot dv/dt$
 - c. $v = \lambda f$
 - d. $F= m \cdot de/dt$
4. What is the unit of Resistivity
 - a. ohm-meter
 - b. ohm
 - c. moh
 - d. unit less
5. For perfect black body
 - a. Amount of heat absorbed by surface in given time = amount of incident on surface in same time
 - b. Amount of heat absorbed by surface in given time > amount of incident on surface in same time
 - c. Amount of heat absorbed by surface in given time < amount of incident on surface in same time
 - d. Not any of above

CHAROTAR UNIVERSITY OF SCI. & TECHNOLOGY CHARUSAT CAMPUS CHANGA	
Examination Semester	_____
Subject	_____ No. _____
Candidate's Seat No.	_____
Section	_____
Date	_____
Sr. Supervisor's Sign.	

6. If frequency of emitted X-ray $\nu = 2 \times 10^{17}$ Hz. Calculate the X-ray wavelength.

- a. 1×10^{-8} m
- b. 1.5×10^{-9} m
- c. 2×10^7 m
- d. 1.5×10^9 m

7. The wavelength range of IR is

- a. 4nm-2mm
- b. $1\text{\AA}$ -5nm
- c. 760nm-1mm
- d. 1nm-2 μ m

8. A person produces sound and this sound is reflected back from an obstacle at a distance x.

The velocity of sound $v=360$ m/s. the time taken by the sound to travel to the obstacle and back is $t=0.1$. Calculate the distance between the person and obstacle x.

- a. 18m
- b. 24m
- c. 30m
- d. 20m

9. The meta stable state life time is

- a. $>10^{-8}$ s
- b. $<10^{-8}$ s
- c. $>10^{-8}$ s
- d. $<10^{-8}$ s

10. What is the frequency of infrasonic waves?

- a. <2 Hz
- b. >20 Hz
- c. $=2$ Hz
- d. <20 Hz

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Bachelor of Physiotherapy (B.P.T)

Semester – II Examination – JUNE 2011

PY102 FUNDAMENTALS OF BIOMEDICAL PHYSICS

Date: 16.06.11, Thursday

Time: 10.20a.m to 1.00p.m

Maximum Marks: 60

Instructions:

1. There are two parts in the paper. **PART –I** contains 10 Questions and **PART – II** contains 4 Questions.
2. **PART –I** should be answered in MCQ 'S question paper & **PART – II** should be answered in two separate answer books which has two sections.
3. Marks on the right side suggest the total marks of that question.
4. Maximum time allotted for **PART – II** is 2 Hr 40min.
5. Draw the figure wherever necessary.

PART-II

SECTION –I

Q.1. Long Question (1 out of 2)

(1× 10= 10)

1. Write notes on

- (i) Pulleys
- (ii) Wheel and Axle
- (iii) Lever

OR

2. Give a detail explanation of X-ray.

Q.2. Write short notes on: (4 out of 5)

(4× 5 = 20)

1. Explain Laplace correction for sound velocity.
2. Write note on Newton's law of cooling
3. Explain shortwave diathermy.
4. Give the application of ultrasonic waves
5. Write a note on friction

SECTION –II

Q.3. Long Question (1 out of 2)

(1× 10= 10)

1. Explain basic principle of fiber optic. Derive formula of numerical aperture and acceptance angle.

OR

2 What is rectifier? Explain half wave rectifier, full wave rectifier and bridge rectifier.

Q.4. Write short notes on: (4 out of 5)

(4 × 5 = 20)

1. A laser source emits a light of wavelength $0.621\mu\text{m}$ and has an output of 35mW .
Calculate how many photons are emitted per minute by this laser source.
2. Give the definition of
 - (i) Refraction
 - (ii) Frequency
 - (iii) Reflection
 - (iv) Magnetization
 - (v) Ohm's law
3. Explain different types of magnetic materials.
4. Give a detail explanation on Induction coil.
5. Write note on PN junction and explain characteristics of PN junction diode.
